

Townsend 6.5-6.11

Questions for Gallicchio

For the double slit experiment (I'm still secretly hoping for local hidden variable theory and don't like quantum stuff haha) - how do we know that it's not the measurement device interacting with the particles that makes it into two beams instead of interference fringes?

Momentum Space

Momentum eigenstates

$$\hat{p}_x |p\rangle = p |p\rangle$$

Wavefunction in momentum

$$\langle p | \psi \rangle$$

$$\langle x | \hat{p}_x | p \rangle = p \langle x | p \rangle = \frac{\hbar}{i} \frac{\partial}{\partial x} \langle x | p \rangle$$

$$|p\rangle = \int dx |x\rangle \langle x | p \rangle$$

DeBroglie wavelength

$$p \frac{\lambda}{h} = 2\pi$$
$$\lambda = \frac{h}{p}$$

$$\langle p | \psi \rangle = \int dx \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{-ipx}{\hbar}} \langle x | \psi \rangle$$

$$\langle x | \psi \rangle = \int dp \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{ipx}{\hbar}} \langle p | \psi \rangle$$

6.6 Gaussian Wave Packets

$$\int dx \psi^*(x) \psi(x) = N^* N \int_{-\infty}^{\infty} dx e^{\frac{-x^2}{a^2}} = 1$$
$$\Rightarrow N = \frac{1}{\sqrt{\sqrt{\pi}a}}$$

EXAMPLE 6.3 Calculate $\langle p_x \rangle$ using the position-space Gaussian wave function and compare your result with (6.70).

SOLUTION

$$\langle p_x \rangle = \langle \psi | \hat{p}_x | \psi \rangle = \int dx \psi^*(x) \frac{\hbar}{i} \frac{\partial}{\partial x} \psi(x)$$
$$= -\frac{\hbar}{ia^2} \int_{-\infty}^{\infty} dx \frac{1}{\sqrt{\pi}a} e^{-x^2/a^2} x = 0$$

Similarly, you can calculate $\langle p_x^2 \rangle$ directly in position space and compare your result with (6.71).

6.7 Double Slit

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$